

Monomials

NoiseModel

$$f(q) = aq^2$$

$$\text{variance gap} = 0$$

$$\text{MSE gap} = -\frac{4\Delta^4 a^2}{\epsilon^4}$$

$$f(q) = aq^3$$

$$\text{variance gap} = \frac{-216\Delta^6 a^2 - 72\Delta^4 a^2 \epsilon^2 q^2}{\epsilon^6}$$

$$\text{MSE gap} = \frac{-216\Delta^6 a^2 - 108\Delta^4 a^2 \epsilon^2 q^2}{\epsilon^6}$$

$$f(q) = aq^4$$

$$\text{variance gap} = \frac{-13248\Delta^8 a^2 - 6336\Delta^6 a^2 \epsilon^2 q^2 - 384\Delta^4 a^2 \epsilon^4 q^4}{\epsilon^8}$$

$$\text{MSE gap} = \frac{-13824\Delta^8 a^2 - 6912\Delta^6 a^2 \epsilon^2 q^2 - 528\Delta^4 a^2 \epsilon^4 q^4}{\epsilon^8}$$

$$f(q) = aq^5$$

$$\begin{aligned} & \text{variance gap} \\ &= \frac{-1324800\Delta^{10} a^2 - 648000\Delta^8 a^2 \epsilon^2 q^2 - 50400\Delta^6 a^2 \epsilon^4 q^4 - 1200\Delta^4 a^2 \epsilon^6 q^6}{\epsilon^{10}} \end{aligned}$$

$$\begin{aligned} & \text{MSE gap} \\ &= \frac{-1324800\Delta^{10} a^2 - 662400\Delta^8 a^2 \epsilon^2 q^2 - 55200\Delta^6 a^2 \epsilon^4 q^4 - 1600\Delta^4 a^2 \epsilon^6 q^6}{\epsilon^{10}} \end{aligned}$$

$$f(q) = aq^6$$

variance gap

$$= \frac{-180921600\Delta^{12}a^2 - 90201600\Delta^{10}a^2\epsilon^2q^2 - 7387200\Delta^8a^2\epsilon^4q^4 - 228960\Delta^6a^2\epsilon^6q^6 - 2880\Delta^4a^2\epsilon^8q^8}{\epsilon^{12}}$$

MSE gap

$$= \frac{-181440000\Delta^{12}a^2 - 90720000\Delta^{10}a^2\epsilon^2q^2 - 7560000\Delta^8a^2\epsilon^4q^4 - 250560\Delta^6a^2\epsilon^6q^6 - 3780\Delta^4a^2\epsilon^8q^8}{\epsilon^{12}}$$

$$f(q) = aq^7$$

variance gap

$$= \frac{-33834931200\Delta^{14}a^2 - 16892064000\Delta^{12}a^2\epsilon^2q^2 - 1401321600\Delta^{10}a^2\epsilon^4q^4 - 45864000\Delta^8a^2\epsilon^6q^6 - 758520\Delta^6a^2\epsilon^8q^8}{\epsilon^{14}}$$

MSE gap

$$= \frac{-33834931200\Delta^{14}a^2 - 16917465600\Delta^{12}a^2\epsilon^2q^2 - 1409788800\Delta^{10}a^2\epsilon^4q^4 - 46992960\Delta^8a^2\epsilon^6q^6 - 829080\Delta^6a^2\epsilon^8q^8}{\epsilon^{14}}$$

$$f(q) = aq^8$$

variance gap

$$= \frac{-8260193894400\Delta^{16}a^2 - 4129284096000\Delta^{14}a^2\epsilon^2q^2 - 343700582400\Delta^{12}a^2\epsilon^4q^4 - 11402496000\Delta^{10}a^2\epsilon^6q^6 - 194400\Delta^8a^2\epsilon^8q^8}{\epsilon^{16}}$$

MSE gap

$$= \frac{-8261819596800\Delta^{16}a^2 - 4130909798400\Delta^{14}a^2\epsilon^2q^2 - 344242483200\Delta^{12}a^2\epsilon^4q^4 - 11474749440\Delta^{10}a^2\epsilon^6q^6 - 201600\Delta^8a^2\epsilon^8q^8}{\epsilon^{16}}$$

$$f(q) = aq^9$$

variance gap

$$= \frac{-2560949482291200\Delta^{18}a^2 - 1280343059251200\Delta^{16}a^2\epsilon^2q^2 - 106662334464000\Delta^{14}a^2\epsilon^4q^4 - 3551021752320\Delta^{12}a^2\epsilon^6q^6 - 518400\Delta^{10}a^2\epsilon^8q^8}{\epsilon^{18}}$$

MSE gap

$$= \frac{-2560949482291200\Delta^{18}a^2 - 1280474741145600\Delta^{16}a^2\epsilon^2q^2 - 106706228428800\Delta^{14}a^2\epsilon^4q^4 - 3556874280960\Delta^{12}a^2\epsilon^6q^6 - 518400\Delta^{10}a^2\epsilon^8q^8}{\epsilon^{18}}$$

$$f(q) = aq^{10}$$

variance gap

$$= \frac{-982939500748800000\Delta^{20}a^2 - 491463166279680000\Delta^{18}a^2\epsilon^2q^2 - 40951971809280000\Delta^{16}a^2\epsilon^4q^4 - 136462678}{}$$

MSE gap

$$= \frac{-982952668938240000\Delta^{20}a^2 - 491476334469120000\Delta^{18}a^2\epsilon^2q^2 - 40956361205760000\Delta^{16}a^2\epsilon^4q^4 - 136521204}{}$$

$$f(q) = aq^{11}$$

variance gap

$$= \frac{-457769719959552000000\Delta^{22}a^2 - 228883266628853760000\Delta^{20}a^2\epsilon^2q^2 - 19073207214673920000\Delta^{18}a^2\epsilon^4q^4 -}{}$$

MSE gap

$$= \frac{-457769719959552000000\Delta^{22}a^2 - 228884859979776000000\Delta^{20}a^2\epsilon^2q^2 - 19073738331648000000\Delta^{18}a^2\epsilon^4q^4 -}{}$$

$$f(q) = aq^{12}$$

variance gap

$$= \frac{-254345078100285849600000\Delta^{24}a^2 - 127172424328876523520000\Delta^{22}a^2\epsilon^2q^2 - 10597644666773176320000\Delta^{20}a^2\epsilon^4q^4 -}{}$$

MSE gap

$$= \frac{-254345307542818652160000\Delta^{24}a^2 - 127172653771409326080000\Delta^{22}a^2\epsilon^2q^2 - 10597721147617443840000\Delta^{20}a^2\epsilon^4q^4 -}{}$$

$$f(q) = aq^{13}$$

variance gap

$$= \frac{-166226219629574844579840000\Delta^{26}a^2 - 83113071038999378657280000\Delta^{24}a^2\epsilon^2q^2 - 69260795593029373132}{}$$

MSE gap

$$= \frac{-166226219629574844579840000\Delta^{26}a^2 - 83113109814787422289920000\Delta^{24}a^2\epsilon^2q^2 - 69260924845656185241}{}$$

$$f(q) = aq^{14}$$

variance gap

$$= \frac{-126246351391018171611217920000\Delta^{28}a^2 - 63123171895481857529610240000\Delta^{26}a^2\epsilon^2q^2 - 52602624246098}{}$$

MSE gap

$$= \frac{-126246358991072628163215360000\Delta^{28}a^2 - 63123179495536314081607680000\Delta^{26}a^2\epsilon^2q^2 - 52602649579613}{}$$

$$f(q) = aq^{15}$$

variance gap

$$= \frac{-110267951301236512881377280000000\Delta^{30}a^2 - 55133973940606003716489216000000\Delta^{28}a^2\epsilon^2q^2 - 45944974}{}$$

MSE gap

$$= \frac{-110267951301236512881377280000000\Delta^{30}a^2 - 55133975650618256440688640000000\Delta^{28}a^2\epsilon^2q^2 - 45944979}{}$$

$$f(q) = aq^{16}$$

variance gap

$$= \frac{-109759803622453853083146387456000000\Delta^{32}a^2 - 54879901592345358192875667456000000\Delta^{30}a^2\epsilon^2q^2 - 45}{}$$

MSE gap

$$= \frac{-10975980406021698978054144000000000\Delta^{32}a^2 - 54879902030108494890270720000000000\Delta^{30}a^2\epsilon^2q^2 - 45}{}$$

$$f(q) = aq^{17}$$

variance gap

$$= \frac{-123518707711584137128817952030720000000\Delta^{34}a^2 - 61759353729278522058861805830144000000\Delta^{32}a^2\epsilon^2q^2 - 45}{}$$

MSE gap

$$= \frac{-123518707711584137128817952030720000000\Delta^{34}a^2 - 61759353855792068564408976015360000000\Delta^{32}a^2\epsilon^2q^2 - 45}{}$$

$$f(q) = aq^{18}$$

variance gap

$$= \frac{-156043953924124017740207680175210496000000\Delta^{36}a^2 - 78021976941566814336205198517600256000000\Delta^{37}a^2 - 3901098847083215816810266812177600128000000\Delta^{38}a^2 - 195054942354160790840513280640000000\Delta^{39}a^2 - 975274711770803954202566400000000\Delta^{40}a^2 - 4876373558854019771012832000000000\Delta^{41}a^2 - 24381867794270098855064160000000000\Delta^{42}a^2 - 121909338971350494275320800000000000\Delta^{43}a^2 - 609546694856752471376604000000000000\Delta^{44}a^2 - 3047733474283762356883020000000000000\Delta^{45}a^2 - 15238667371418811784415100000000000000\Delta^{46}a^2 - 76193336857094058922075500000000000000\Delta^{47}a^2 - 380966684285470294610377500000000000000\Delta^{48}a^2 - 1904833421427351473051887500000000000000\Delta^{49}a^2 - 9524167107136757365259437500000000000000\Delta^{50}a^2 - 47620835535683786826297187500000000000000\Delta^{51}a^2 - 238104177678418934131485937500000000000000\Delta^{52}a^2 - 1190520888392094670657429687500000000000000\Delta^{53}a^2 - 5952604441960473353287148437500000000000000\Delta^{54}a^2 - 29763022209802366766435742187500000000000000\Delta^{55}a^2 - 148815111049011833832218710937500000000000000\Delta^{56}a^2 - 744075555245059169161093554687500000000000000\Delta^{57}a^2 - 3720377776225295845805467773437500000000000000\Delta^{58}a^2 - 18601888881126479229027338867187500000000000000\Delta^{59}a^2 - 93009444405632396145136694285937500000000000000\Delta^{60}a^2 - 465047222028161980725683471429687500000000000000\Delta^{61}a^2 - 2325236110140809903628417357148437500000000000000\Delta^{62}a^2 - 11626180550704049518142086937148437500000000000000\Delta^{63}a^2 - 58130902753520247590710434685714843750000000000000\Delta^{64}a^2 - 290654513767601237953552173428571484375000000000000\Delta^{65}a^2 - 1453272568838006189767760867142857148437500000000000\Delta^{66}a^2 - 7266362844190030948838804335714285714843750000000000\Delta^{67}a^2 - 36331814220950154744194021678571428571484375000000000\Delta^{68}a^2 - 181659071104750773720970108392857142857148437500000000\Delta^{69}a^2 - 908295355523753868604850541964285714285714843750000000\Delta^{70}a^2 - 4541476777618769343024252709821428571428571484375000000\Delta^{71}a^2 - 227073838880938467151212635491071428571428571484375000000\Delta^{72}a^2 - 1135369194404692335756063177455357142857142857148437500000\Delta^{73}a^2 - 5676845972023461678780315887276785714285714285714843750000\Delta^{74}a^2 - 28384229860117308393901579436383928571428571428571484375000\Delta^{75}a^2 - 14192114930058654196950789718191964285714285714285714843750\Delta^{76}a^2 - 70960574650293270984753948590959821428571428571428571484375\Delta^{77}a^2 - 3548028732514663549237697429547991071428571428571428571484375\Delta^{78}a^2 - 177401436625733177461884871477399553571428571428571428571484375\Delta^{79}a^2 - 8870071831286658873094243573869977678571428571428571428571484375\Delta^{80}a^2 - 443503591564332943654712178693498883928571428571428571428571484375\Delta^{81}a^2 - 22175179578216647182735608934674944196428571428571428571428571484375\Delta^{82}a^2 - 1108758978910832359136780446733747209821428571428571428571428571484375\Delta^{83}a^2 - 55437948945541617956839022336687360491071428571428571428571428571484375\Delta^{84}a^2 - 2771897447277080897841951116834368024553571428571428571428571428571484375\Delta^{85}a^2 - 138594872363854044892097555841718401227678571428571428571428571428571484375\Delta^{86}a^2 - 6929743618192702244604877792085920061383928571428571428571428571428571484375\Delta^{87}a^2 - 346487180909635112230243889604296003069196428571428571428571428571428571484375\Delta^{88}a^2 - 17324359045481755611512194480214800153459821428571428571428571428571428571484375\Delta^{89}a^2 - 866217952274087780575609724010740007672991071428571428571428571428571428571484375\Delta^{90}a^2 - 43310897613704389028780486200537000383649553571428571428571428571428571428571484375\Delta^{91}a^2 - 2165544880685219451439024310026850019182477678571428571428571428571428571428571484375\Delta^{92}a^2 - 108277244034260972571951215501342500959123883928571428571428571428571428571428571484375\Delta^{93}a^2 - 5413862201713048628597560775067125047956194196428571428571428571428571428571428571484375\Delta^{94}a^2 - 270693110085652431429878038753356252397809709821428571428571428571428571428571428571484375\Delta^{95}a^2 - 13534655504282621571493901937667812619890485491071428571428571428571428571428571428571484375\Delta^{96}a^2 - 676732775214131078574695096883390630994524274553571428571428571428571428571428571428571484375\Delta^{97}a^2 - 33836638760706553928734754844169531549726213727678571428571428571428571428571428571428571484375\Delta^{98}a^2 - 1691831938035327696436737742208476577486310686383928571428571428571428571428571428571428571484375\Delta^{99}a^2 - 84591596901766384821836887110423828874315534319196428571428571428571428571428571428571428571484375\Delta^{100}a^2 - 4229579845088319241091694355521191193715776715959821428571428571428571428571428571428571428571484375\Delta^{101}a^2 - 211478992254415962054584717776059559685788835797991071428571428571428571428571428571428571428571484375\Delta^{102}a^2 - 10573949612720798102729235888802977984289441789899553571428571428571428571428571428571428571428571484375\Delta^{103}a^2 - 528697480636039905136146194440148899214472089494977678571428571428571428571428571428571428571428571484375\Delta^{104}a^2 - 26434874031801995256807309722007444960723604474748883928571428571428571428571428571428571428571428571484375\Delta^{105}a^2 - 1321743701590099762840365486100372248036180223737444196428571428571428571428571428571428571428571428571484375\Delta^{106}a^2 - 66087185079504988142018274305018612401809011186872209821428571428571428571428571428571428571428571428571484375\Delta^{107}a^2 - 3304359253975249407100913715250930620090450559343611491071428571428571428571428571428571428571428571428571484375\Delta^{108}a^2 - 165217962698762470355045685762546531004522527967180574553571428571428571428571428571428571428571428571428571484375\Delta^{109}a^2 - 8260898134938123517752284288127326550226126398359028727678571428571428571428571428571428571428571428571428571484375\Delta^{110}a^2 - 413044906746906175887614214406366327511306319917951436383928571428571428571428571428571428571428571428571428571484375\Delta^{111}a^2 - 20652245337345308794380710720318316375565315995897571819196428571428571428571428571428571428571428571428571484375\Delta^{112}a^2 - 1032612266867265439719035536015915818778265799794878590959821428571428571428571428571428571428571428571428571428571484375\Delta^{113}a^2 - 51630613343363271985951776800795790938913289989743929547991071428571428571428571428571428571428571428571428571428571484375\Delta^{114}a^2 - 2581530667168163599297588840039789546945664499487196477399553571428571428571428571428571428571428571428571428571428571484375\Delta^{115}a^2 - 129076533358408179964879442001989477347283224972359823869977678571428571428571428571428571428571428571428571428571428571484375\Delta^{116}a^2 - 6453826667920408998243972100099473867364161248617991193498883928571428571428571428571428571428571428571428571428571428571484375\Delta^{117}a^2 - 322691333396020449912198605004973693368208062430899559674944196428571428571428571428571428571428571428571428571428571428571484375\Delta^{118}a^2 - 16134566669801022495609930250248684668410403121544977983747209821428571428571428571428571428571428571428571428571428571428571484375\Delta^{119}a^2 - 806728333490051124780496512512434233420520156077248899187360491071428571428571428571428571428571428571428571428571428571428571484375\Delta^{120}a^2 - 40336416674502556239024825625621711671026007803862444959368024553571428571428571428571428571428571428571428571428571428571428571484375\Delta^{121}a^2 - 2016820833725127811951241281281085583551300390193122247968401227678571428571428571428571428571428571428571428571428571428571428571484375\Delta^{122}a^2 - 100841041686256390597562064064054279177565019509656112398420061383928571428571428571428571428571428571428571428571428571428571428571484375\Delta^{123}a^2 - 5042052084312819529878103203202713958878250975482805619921003069196428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{124}a^2 - 252102604215640976493905160160135697943912548774140280996050153459821428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{125}a^2 - 12605130210782048824695258008006784897195627438707014049802507673991071428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{126}a^2 - 630256510539102441234762900400339244859781371935350702490125383699553571428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{127}a^2 - 31512825526955122061738145020016962242989068596767535124506269184977678571428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{128}a^2 - 1575641276347756103086907251000848112149453429838376756225313459248883928571428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{129}a^2 - 78782063817387805154345362550042405607472671491918837811265672962444196428571428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{130}a^2 - 3939103190869390257717268127502120280373633574595941890563283648122209821428571428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{131}a^2 - 196955159543469512885863406375106014018681678729797094528164182406110491071428571428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{132}a^2 - 9847757977173475644293170318755300700934083936489854726408209120305524553571428571428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{133}a^2 - 492387898858673782214658515937765035046704196824492736320410456015276227678571428571428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{134}a^2 - 24619394942933689110732925796888251752335209841224636816020522800763811383928571428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{135}a^2 - 1230969747146684455536646289844412587616760492061231840801026140038190569196428571428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{136}a^2 - 61548487357334222776832314492220629380838024603061592040051307001909528459821428571428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{137}a^2 - 3077424367866711138841615724611031469041901230153079602002565350095476422991071428571428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{138}a^2 - 153871218393335556942080786230551573452095061507653980100128267504773821149553571428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{139}a^2 - 7693560919666777847104039311527578672604753075382699005006413375238691057477678571428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{140}a^2 - 384678045983338892355201965576378933630237653769134950250320668761934552873883928571428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{141}a^2 - 19233902299166944617760098278818946681511882688456747512516033438096727643694196428571428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{142}a^2 - 961695114958347230888004913940947334075594134422837375625801671904836382184709821428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{143}a^2 - 48084755747917361544400245697047366703779706721141868781290083595241819109235491071428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{144}a^2 - 2404237787395868077220012284852368335188985336057093439064504179762090955461774553571428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{145}a^2 - 120211889369793403861000614242618416759449266802854671953225208988104547773088727678571428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{146}a^2 - 6010594468489670193050030712130920837972463340142733597661260449405227388654436383928571428571428571428571428571428571428571428571428571428571428571484375\Delta^{147}a^2 - 300529723424483509652501535606546041898623167007136679883063022470261369432721819196428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{148}a^2 - 15026486171224175482625076780327302094931158350356833994153151123513068471636090959821428571428571428571428571428571428571428571428571428571428571428571484375\Delta^{149}a^2 - 751324308561208774131253839016365104746557917501784199707657556175653423581804547991071428571428571428571428571428571428571428571428571428571428571484375\Delta^{150}a^2 - 37566215428060438706562691950818255237327895875089209985382877808782671179090227399553571428571428571428571428571428571428571428571428571428571428571484375\Delta^{151}a^2 - 1878310771403021935328134597540912761866394793754460499269143890439133558954511369977678571428571428571428571428571428571428571428571428571428571428571484375\Delta^{152}a^2 - 9391553857015109676640672987704563809331973968772302$$

$$\text{variance gap} = -300a^2q^6\sigma^4 - 2250a^2q^4\sigma^6 - 3900a^2q^2\sigma^8 - 825a^2\sigma^{10}$$

$$\text{MSE gap} = -400a^2q^6\sigma^4 - 2550a^2q^4\sigma^6 - 4125a^2q^2\sigma^8 - 825a^2\sigma^{10}$$

$$f(q) = aq^6$$

$$\begin{aligned} \text{variance gap} = & -720a^2q^8\sigma^4 - 10080a^2q^6\sigma^6 - 44100a^2q^4\sigma^8 \\ & - 56700a^2q^2\sigma^{10} - 9450a^2\sigma^{12} \end{aligned}$$

$$\begin{aligned} \text{MSE gap} = & -945a^2q^8\sigma^4 - 11430a^2q^6\sigma^6 - 46575a^2q^4\sigma^8 \\ & - 58050a^2q^2\sigma^{10} - 9675a^2\sigma^{12} \end{aligned}$$

$$f(q) = aq^7$$

$$\begin{aligned} \text{variance gap} = & -1470a^2q^{10}\sigma^4 - 33075a^2q^8\sigma^6 - 270480a^2q^6\sigma^8 \\ & - 870975a^2q^4\sigma^{10} - 899640a^2q^2\sigma^{12} - 130095a^2\sigma^{14} \end{aligned}$$

$$\begin{aligned} \text{MSE gap} = & -1911a^2q^{10}\sigma^4 - 37485a^2q^8\sigma^6 - 285915a^2q^6\sigma^8 \\ & - 893025a^2q^4\sigma^{10} - 910665a^2q^2\sigma^{12} - 130095a^2\sigma^{14} \end{aligned}$$

$$f(q) = aq^8$$

$$\begin{aligned} \text{variance gap} = & -2688a^2q^{12}\sigma^4 - 88704a^2q^{10}\sigma^6 - 1165920a^2q^8\sigma^8 \\ & - 7008960a^2q^6\sigma^{10} - 18133920a^2q^4\sigma^{12} \\ & - 15805440a^2q^2\sigma^{14} - 1975680a^2\sigma^{16} \end{aligned}$$

$$\begin{aligned} \text{MSE gap} = & -3472a^2q^{12}\sigma^4 - 100464a^2q^{10}\sigma^6 - 1233540a^2q^8\sigma^8 - 7191240a^2q^6\sigma^{10} \\ & - 18354420a^2q^4\sigma^{12} - 15893640a^2q^2\sigma^{14} - 1986705a^2\sigma^{16} \end{aligned}$$

$$f(q) = aq^9$$

$$\begin{aligned} \text{variance gap} = & -4536a^2q^{14}\sigma^4 - 206388a^2q^{12}\sigma^6 - 3978072a^2q^{10}\sigma^8 \\ & - 38425590a^2q^8\sigma^{10} - 185590440a^2q^6\sigma^{12} \\ & - 404599860a^2q^4\sigma^{14} - 305975880a^2q^2\sigma^{16} - 34096545a^2\sigma^{18} \end{aligned}$$

$$\begin{aligned} \text{MSE gap} = & -5832a^2q^{14}\sigma^4 - 233604a^2q^{12}\sigma^6 - 4211676a^2q^{10}\sigma^8 \\ & - 39446190a^2q^8\sigma^{10} - 187892460a^2q^6\sigma^{12} \\ & - 406981260a^2q^4\sigma^{14} - 306868905a^2q^2\sigma^{16} - 34096545a^2\sigma^{18} \end{aligned}$$

$$f(q) = aq^{10}$$

$$\begin{aligned} \text{variance gap} = & -7200a^2q^{16}\sigma^4 - 432000a^2q^{14}\sigma^6 - 11478600a^2q^{12}\sigma^8 \\ & - 162577800a^2q^{10}\sigma^{10} - 1261745100a^2q^8\sigma^{12} - 5134298400a^2q^6\sigma^{14} \\ & - 9711009000a^2q^4\sigma^{16} - 6502072500a^2q^2\sigma^{18} - 650207250a^2\sigma^{20} \end{aligned}$$

$$\begin{aligned} \text{MSE gap} = & -9225a^2q^{16}\sigma^4 - 488700a^2q^{14}\sigma^6 - 12159000a^2q^{12}\sigma^8 \\ & - 166972050a^2q^{10}\sigma^{10} - 1277706150a^2q^8\sigma^{12} - 5165256600a^2q^6\sigma^{14} \\ & - 9739288125a^2q^4\sigma^{16} - 6511002750a^2q^2\sigma^{18} - 651100275a^2\sigma^{20} \end{aligned}$$

$$f(q) = aq^{11}$$

$$\begin{aligned} \text{variance gap} = & -10890a^2q^{18}\sigma^4 - 833085a^2q^{16}\sigma^6 \\ & - 29185200a^2q^{14}\sigma^8 - 569819250a^2q^{12}\sigma^{10} \\ & - 6484733640a^2q^{10}\sigma^{12} - 42402556350a^2q^8\sigma^{14} \\ & - 149700474000a^2q^6\sigma^{16} - 250612795125a^2q^4\sigma^{18} \\ & - 150695275500a^2q^2\sigma^{20} - 13709393775a^2\sigma^{22} \end{aligned}$$

$$\begin{aligned} \text{MSE gap} = & -13915a^2q^{18}\sigma^4 - 941985a^2q^{16}\sigma^6 - 30927600a^2q^{14}\sigma^8 \\ & - 585446400a^2q^{12}\sigma^{10} - 6568205490a^2q^{10}\sigma^{12} - 42663262950a^2q^8\sigma^{14} \\ & - 150144704325a^2q^6\sigma^{16} - 250972981875a^2q^4\sigma^{18} \\ & - 150803331525a^2q^2\sigma^{20} - 13709393775a^2\sigma^{22} \end{aligned}$$

$$f(q) = aq^{12}$$

$$\begin{aligned} \text{variance gap} = & -15840a^2q^{20}\sigma^4 - 1504800a^2q^{18}\sigma^6 - 67205160a^2q^{16}\sigma^8 \\ & - 1729965600a^2q^{14}\sigma^{10} - 27140263920a^2q^{12}\sigma^{12} \\ & - 260245560960a^2q^{10}\sigma^{14} - 1476477525600a^2q^8\sigma^{16} \\ & - 4613765864400a^2q^6\sigma^{18} - 6936373521000a^2q^4\sigma^{20} \\ & - 3787765027200a^2q^2\sigma^{22} - 315647085600a^2\sigma^{24} \end{aligned}$$

$$\begin{aligned} \text{MSE gap} = & -20196a^2q^{20}\sigma^4 - 1700820a^2q^{18}\sigma^6 - 71239905a^2q^{16}\sigma^8 \\ & - 1777990500a^2q^{14}\sigma^{10} - 27494962110a^2q^{12}\sigma^{12} \\ & - 261872919000a^2q^{10}\sigma^{14} - 1480938695775a^2q^8\sigma^{16} \\ & - 4620537375300a^2q^6\sigma^{18} - 6941344098150a^2q^4\sigma^{20} \\ & - 3789061699500a^2q^2\sigma^{22} - 315755141625a^2\sigma^{24} \end{aligned}$$

$$f(q) = aq^{13}$$

$$\begin{aligned}
\text{variance gap} = & -22308a^2q^{22}\sigma^4 - 2576574a^2q^{20}\sigma^6 \\
& - 142882740a^2q^{18}\sigma^8 - 4688779095a^2q^{16}\sigma^{10} \\
& - 96986930040a^2q^{14}\sigma^{12} - 1282114934100a^2q^{12}\sigma^{14} \\
& - 10667480143320a^2q^{10}\sigma^{16} - 53569582241175a^2q^8\sigma^{18} \\
& - 150331884803100a^2q^6\sigma^{20} - 205236293412150a^2q^4\sigma^{22} \\
& - 102676883809500a^2q^2\sigma^{24} - 7899626559825a^2\sigma^{26}
\end{aligned}$$

$$\begin{aligned}
\text{MSE gap} = & -28392a^2q^{22}\sigma^4 - 2911194a^2q^{20}\sigma^6 \\
& - 151499205a^2q^{18}\sigma^8 - 4820284755a^2q^{16}\sigma^{10} \\
& - 98271368910a^2q^{14}\sigma^{12} - 1290252223260a^2q^{12}\sigma^{14} \\
& - 10700234840295a^2q^{10}\sigma^{16} - 53649584863875a^2q^8\sigma^{18} \\
& - 150441453612450a^2q^6\sigma^{20} - 205309339285050a^2q^4\sigma^{22} \\
& - 102695145277725a^2q^2\sigma^{24} - 7899626559825a^2\sigma^{26}
\end{aligned}$$

$$f(q) = aq^{14}$$

$$\begin{aligned}
\text{variance gap} = & -30576a^2q^{24}\sigma^4 - 4219488a^2q^{22}\sigma^6 - 284456172a^2q^{20}\sigma^8 \\
& - 11590558980a^2q^{18}\sigma^{10} - 305644288950a^2q^{16}\sigma^{12} \\
& - 5327531969760a^2q^{14}\sigma^{14} - 61111702131480a^2q^{12}\sigma^{16} \\
& - 450077827219020a^2q^{10}\sigma^{18} - 2029827122873550a^2q^8\sigma^{20} \\
& - 5172751605621600a^2q^6\sigma^{22} - 6469776997784100a^2q^4\sigma^{24} \\
& - 2986936496844300a^2q^2\sigma^{26} - 213352606917450a^2\sigma^{28}
\end{aligned}$$

$$\begin{aligned}
\text{MSE gap} = & -38857a^2q^{24}\sigma^4 - 4766034a^2q^{22}\sigma^6 - 301672371a^2q^{20}\sigma^8 \\
& - 11918486580a^2q^{18}\sigma^{10} - 309739284855a^2q^{16}\sigma^{12} \\
& - 5361792205770a^2q^{14}\sigma^{14} - 61302051805995a^2q^{12}\sigma^{16} \\
& - 450760400320230a^2q^{10}\sigma^{18} - 2031330650424075a^2q^8\sigma^{20} \\
& - 5174626449692700a^2q^6\sigma^{22} - 6470927470282275a^2q^4\sigma^{24} \\
& - 2987192157399450a^2q^2\sigma^{26} - 213370868385675a^2\sigma^{28}
\end{aligned}$$

$$f(q) = aq^{15}$$

$$\begin{aligned}
\text{variance gap} = & -40950a^2q^{26}\sigma^4 - 6654375a^2q^{24}\sigma^6 \\
& - 535953600a^2q^{22}\sigma^8 - 26547946425a^2q^{20}\sigma^{10} \\
& - 869285617200a^2q^{18}\sigma^{12} - 19317298925925a^2q^{16}\sigma^{14} \\
& - 292183978086000a^2q^{14}\sigma^{16} - 2966854968952875a^2q^{12}\sigma^{18} \\
& - 19623871149082200a^2q^{10}\sigma^{20} - 80369726038977375a^2q^8\sigma^{22} \\
& - 187638888063627000a^2q^6\sigma^{24} - 216572089931071875a^2q^4\sigma^{26} \\
& - 92830526358570000a^2q^2\sigma^{28} - 6188975679261375a^2\sigma^{30}
\end{aligned}$$

$$\begin{aligned}
\text{MSE gap} = & -51975a^2q^{26}\sigma^4 - 7514325a^2q^{24}\sigma^6 \\
& - 568488375a^2q^{22}\sigma^8 - 27304702425a^2q^{20}\sigma^{10} \\
& - 881051596425a^2q^{18}\sigma^{12} - 19442986638075a^2q^{16}\sigma^{14} \\
& - 293105777839875a^2q^{14}\sigma^{16} - 2971416646823625a^2q^{12}\sigma^{18} \\
& - 19638620328251925a^2q^{10}\sigma^{20} - 80399309617501875a^2q^8\sigma^{22} \\
& - 187672763087184375a^2q^6\sigma^{24} - 216591264472708125a^2q^4\sigma^{26} \\
& - 92834635188920625a^2q^2\sigma^{28} - 6188975679261375a^2\sigma^{30}
\end{aligned}$$

$$f(q) = aq^{16}$$

$$\begin{aligned}
\text{variance gap} = & -53760a^2q^{28}\sigma^4 - 10160640a^2q^{26}\sigma^6 - 963580800a^2q^{24}\sigma^8 \\
& - 57023366400a^2q^{22}\sigma^{10} - 2269904757120a^2q^{20}\sigma^{12} \\
& - 62635987814400a^2q^{18}\sigma^{14} - 1207871829964800a^2q^{16}\sigma^{16} \\
& - 16172732800224000a^2q^{14}\sigma^{18} - 147488111365795200a^2q^{12}\sigma^{20} \\
& - 885906579779020800a^2q^{10}\sigma^{22} - 3324050106433056000a^2q^8\sigma^{24} \\
& - 7161624209611872000a^2q^6\sigma^{26} - 7674356327798160000a^2q^4\sigma^{28} \\
& - 3069980037476352000a^2q^2\sigma^{30} - 191873752342272000a^2\sigma^{32}
\end{aligned}$$

$$\begin{aligned}
\text{MSE gap} = & -68160a^2q^{28}\sigma^4 - 11471040a^2q^{26}\sigma^6 - 1022221200a^2q^{24}\sigma^8 \\
& - 58659400800a^2q^{22}\sigma^{10} - 2300906527920a^2q^{20}\sigma^{12} \\
& - 63047814429600a^2q^{18}\sigma^{14} - 1211726493677700a^2q^{16}\sigma^{16} \\
& - 16197908288562000a^2q^{14}\sigma^{18} - 147600429356179800a^2q^{12}\sigma^{20} \\
& - 886237234097014800a^2q^{10}\sigma^{22} - 3324658943783677500a^2q^8\sigma^{24} \\
& - 7162268474210850000a^2q^6\sigma^{26} - 7674695991107145000a^2q^4\sigma^{28} \\
& - 3070045778761962000a^2q^2\sigma^{30} - 191877861172622625a^2\sigma^{32}
\end{aligned}$$

$$f(q) = aq^{17}$$

$$\begin{aligned}
\text{variance gap} = & -69360a^2q^{30}\sigma^4 - 15085800a^2q^{28}\sigma^6 - 1663877040a^2q^{26}\sigma^8 \\
& - 115939141500a^2q^{24}\sigma^{10} - 5513820989040a^2q^{22}\sigma^{12} \\
& - 184970541876120a^2q^{20}\sigma^{14} - 4429374118369200a^2q^{18}\sigma^{16} \\
& - 75613762928053350a^2q^{16}\sigma^{18} - 909293665829749200a^2q^{14}\sigma^{20} \\
& - 7530529869597735000a^2q^{12}\sigma^{22} \\
& - 41441162220147690000a^2q^{10}\sigma^{24} \\
& - 143491929707832205500a^2q^8\sigma^{26} \\
& - 287027392309131474000a^2q^6\sigma^{28} \\
& - 287050060985624937000a^2q^4\sigma^{30} \\
& - 107647983664719498000a^2q^2\sigma^{32} - 6332304183334754625a^2\sigma^{34}
\end{aligned}$$

$$\begin{aligned}
\text{MSE gap} = & -87856a^2q^{30}\sigma^4 - 17027880a^2q^{28}\sigma^6 - 1765350720a^2q^{26}\sigma^8 \\
& - 119284374300a^2q^{24}\sigma^{10} - 5589732526560a^2q^{22}\sigma^{12} \\
& - 186198191819640a^2q^{20}\sigma^{14} - 4443656692374900a^2q^{18}\sigma^{16} \\
& - 75732790192809750a^2q^{16}\sigma^{18} - 909994421393857800a^2q^{14}\sigma^{20} \\
& - 7533376129335256200a^2q^{12}\sigma^{22} - 41448853390545701100a^2q^{10}\sigma^{24} \\
& - 143505018067338427500a^2q^8\sigma^{26} \\
& - 287040269566065015000a^2q^6\sigma^{28} \\
& - 287056394062805367000a^2q^4\sigma^{30} \\
& - 107649171116690828625a^2q^2\sigma^{32} - 6332304183334754625a^2\sigma^{34}
\end{aligned}$$

$$f(q) = aq^{18}$$

$$\begin{aligned}
\text{variance gap} = & -88128a^2q^{32}\sigma^4 - 21855744a^2q^{30}\sigma^6 \\
& - 2773938960a^2q^{28}\sigma^8 - 224796131280a^2q^{26}\sigma^{10} \\
& - 12588125691960a^2q^{24}\sigma^{12} - 504530370751680a^2q^{22}\sigma^{14} \\
& - 14688226472839920a^2q^{20}\sigma^{16} - 311370988504375800a^2q^{18}\sigma^{18} \\
& - 4774008865621495500a^2q^{16}\sigma^{20} \\
& - 52136110267210569600a^2q^{14}\sigma^{22} \\
& - 395583844331489814000a^2q^{12}\sigma^{24} \\
& - 2008921377011435686800a^2q^{10}\sigma^{26} \\
& - 6458203227403314387000a^2q^8\sigma^{28} \\
& - 12056243882020602264000a^2q^6\sigma^{30} \\
& - 11303183405266210458000a^2q^4\sigma^{32} \\
& - 3989439101718408838500a^2q^2\sigma^{34} \\
& - 221635505651022713250a^2\sigma^{36}
\end{aligned}$$

$$\begin{aligned}
\text{MSE gap} = & -111537a^2q^{32}\sigma^4 - 24664824a^2q^{30}\sigma^6 \\
& - 2943420120a^2q^{28}\sigma^8 - 231314601420a^2q^{26}\sigma^{10} \\
& - 12762675836820a^2q^{24}\sigma^{12} - 507907449536760a^2q^{22}\sigma^{14} \\
& - 14736035616922620a^2q^{20}\sigma^{16} - 311866130537611650a^2q^{18}\sigma^{18} \\
& - 4777728056406355050a^2q^{16}\sigma^{20} - 52156042820975293200a^2q^{14}\sigma^{22} \\
& - 395658150517959228900a^2q^{12}\sigma^{24} \\
& - 2009106936172822285800a^2q^{10}\sigma^{26} \\
& - 6458496765530627317500a^2q^8\sigma^{28} \\
& - 12056513671108488582000a^2q^6\sigma^{30} \\
& - 11303308087723200173625a^2q^4\sigma^{32} \\
& - 3989460475853892789750a^2q^2\sigma^{34} - 221636693102994043875a^2\sigma^{36}
\end{aligned}$$

$$f(q) = aq^{19}$$

$$\begin{aligned}
\text{variance gap} = & -110466a^2q^{34}\sigma^4 - 30985713a^2q^{32}\sigma^6 \\
& - 4484035872a^2q^{30}\sigma^8 - 418154682420a^2q^{28}\sigma^{10} \\
& - 27233081351280a^2q^{26}\sigma^{12} - 1285226745975900a^2q^{24}\sigma^{14} \\
& - 44703889863141600a^2q^{22}\sigma^{16} - 1152132730953954750a^2q^{20}\sigma^{18} \\
& - 21936266882808478200a^2q^{18}\sigma^{20} \\
& - 305438280920606884050a^2q^{16}\sigma^{22} \\
& - 3056046588066442370400a^2q^{14}\sigma^{24} \\
& - 21398318678847952541700a^2q^{12}\sigma^{26} \\
& - 100892395674802686754800a^2q^{10}\sigma^{28} \\
& - 302699968055644879219500a^2q^8\sigma^{30} \\
& - 529745765476460993460000a^2q^6\sigma^{32} \\
& - 467432315683304961614625a^2q^4\sigma^{34} \\
& - 155812356193050521464500a^2q^2\sigma^{36} \\
& - 8200672887537482727375a^2\sigma^{38}
\end{aligned}$$

$$\begin{aligned}
\text{MSE gap} = & -139707a^2q^{34}\sigma^4 - 34962489a^2q^{32}\sigma^6 - 4758433416a^2q^{30}\sigma^8 \\
& - 430333558920a^2q^{28}\sigma^{10} - 27613131891660a^2q^{26}\sigma^{12} \\
& - 1293895739862180a^2q^{24}\sigma^{14} - 44850646647244620a^2q^{22}\sigma^{16} \\
& - 1153981885161285180a^2q^{20}\sigma^{18} - 21953526207739676550a^2q^{18}\sigma^{20} \\
& - 305556285360710434050a^2q^{16}\sigma^{22} \\
& - 3056626994808389102100a^2q^{14}\sigma^{24} \\
& - 21400318100401928988300a^2q^{12}\sigma^{26} \\
& - 100897036743752821271700a^2q^{10}\sigma^{28} \\
& - 302706826778231284909500a^2q^8\sigma^{30} \\
& - 529751681124691768367625a^2q^6\sigma^{32} \\
& - 467434887704274863748375a^2q^4\sigma^{34} \\
& - 155812784863212171820125a^2q^2\sigma^{36} \\
& - 8200672887537482727375a^2\sigma^{38}
\end{aligned}$$